Transferable Mangrove Extent Mapping from Tanager-1 Hyperspectral Imagery Using Adaptive Spectral Thresholds and Pseudo-Labels

Planet Tanager Open Data Competition 2026 | Technical Memo

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1. Summary and Environmental Context

Mangroves store disproportionately large blue-carbon stocks yet remain difficult to monitor at scale, because conventional multispectral sensors lack the spectral resolution to separate mangrove from adjacent coastal vegetation reliably across geographically diverse sites (Kamal & Phinn, 2011; Lassalle et al., 2023; Rahmila et al., 2026; Yang et al., 2024). This work demonstrates that Tanager-1 hyperspectral imagery, combined with per-scene adaptive spectral thresholds, produces transferable mangrove extent maps across six scenes on four continents without any site-specific training data or field reference samples. A classifier trained on a single scene in Indonesia (Sangatta) and applied zero-shot to five further scenes achieves strong agreement with the Global Mangrove Watch (GMW) v3 reference at ecologically favorable sites (Kappa up to 0.63) while failing predictably at a site with extremely sparse mangrove cover, and the transition between these outcomes is diagnosable directly from each scene's spectral distribution. The framework therefore offers both a practical, reference-free monitoring capability for data-scarce regions and an interpretable account of where that capability holds.

2. Data and Observation Sites

The framework utilized six Tanager-1 scenes featuring 426-band VIS-SWIR surface reflectance at a 30 m resolution. Sangatta in Kutai Timur, Indonesia, functioned as the sole training anchor before the classifier was applied without retraining to five transfer scenes spanning four continents (**Table 1**). To ensure rigorous evaluation, the 2020 release of GMW v3 served exclusively as an independent accuracy reference and never influenced the training or calibration of any model component (Bunting et al., 2018).

Table 1. Six Tanager-1 scenes used in the modeling pipeline, Sangatta as training anchor.

Site	Scene ID	Date	Coastal setting	GMW v3 area (ha)
Sangatta, Indonesia	20250302_030003_92_4001	2 Mar 2025	Tropical estuarine	1,216.4
El Salvador	20250223_165546_32_4001	23 Feb 2025	Wet-dry tropical estuarine	8,892.5
Australia	20250608_014315_58_4001	8 Jun 2025	Wet-dry tropical estuarine	5,797.6
Belize 1	20250824_171853_67_4001	24 Aug 2025	Non-estuarine (barrier reef)	2,730.1
Belize 2	20250824_171857_84_4001	24 Aug 2025	Non-estuarine (barrier reef)	718.8
Gulf of Kutch, Gujarat	20250311_061550_53_4001	11 Mar 2025	Monsoonal semi-arid intertidal	345.0

3. Method

3.1 Workflow

The pipeline is fully deterministic (fixed seed 42) and proceeds through seven stages: (1) extraction of six index-relevant bands from the Tanager HDF5; (2) computation of eight spectral indices; (3) REIP extraction from the full 426-band spectrum; (4) construction of the coastal candidate zone (MNDWI water mask plus distance buffer, capped at 500 m); (5) per-scene adaptive threshold calibration; (6) pseudo-label generation and XGBoost classification on all nine features; and (7) zero-shot transfer to five further scenes, each independently recalibrated at stages 4 and 5 but never retrained. Stages 1 to 3 run in notebook NB01, stages 4 to 6 in NB02, and stage 7 in NB03.

3.2 Spectral indices

The pipeline derived nine features per scene, encompassing standard spectral indices such as the Normalized Difference Vegetation Index (NDVI) (Huete, 1988; Rouse Jr et al., 1974), Modified Normalized Difference Water Index (MNDWI) (Xu, 2006), Normalized Difference Moisture Index (NDMI) (Gao, 1996), Combined Mangrove Recognition Index (CMRI) (Gupta et al., 2018), Normalized Difference Red Edge Index (NDRE) (Gitelson & Merzlyak, 1994), Soil Adjusted Vegetation Index (SAVI) (Huete, 1988), and the Mangrove Vegetation Index (MVI) (Baloloy et al., 2020). To fully exploit Tanager's continuous red-edge and SWIR coverage, the model also incorporated two hyperspectral-specific features, namely the Enhanced Mangrove Index (EMI) (Rahmila et al., 2026) and the Red-edge inflection Point (REIP) (Atzberger et al., 2014; Cho & Skidmore, 2006; Guyot et al., 1988; Hallik et al., 2019). This emphasis on SWIR and red-edge information follows Lassalle et al. (2023), who identify these regions as most informative for discriminating mangrove from surrounding vegetation.

3.3 Adaptive spectral thresholds

Calibration relies on each scene's specific candidate-zone histogram by applying bimodal valley detection for NDMI and a forced-Otsu split for the right-skewed MVI (**Figure 1**). This coastal candidate zone originates from an MNDWI water mask and a distance-transform buffer that is hard-capped at 500 m. The cap prevents inland forests, notably Kutai National Park, from inflating the buffer to the roughly 1,350 m extent derived by Otsu and admitting substantial inland noise. This strictly per-scene calibration ensures that no threshold, band, or parameter is imported from the training site, thereby enabling effective zero shot transfer.

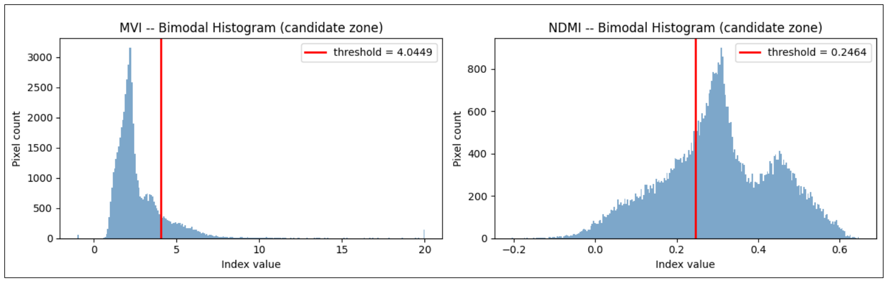

Figure 1. Per-scene adaptive MVI (forced-Otsu split) and NDMI (bimodal valley) threshold histograms for Sangatta within the candidate zone (n = 57,014 pixels; MVI = 4.0449, NDMI = 0.2464).

3.4 Pseudo-labels and classification

Pixels exceeding both the MVI and NDMI thresholds (AND-logic) become positive pseudo-labels. XGBoost serves as the primary classifier and is trained exclusively on the Sangatta scene, utilizing RandomizedSearchCV for tuning (50 iterations, 5-fold cross-validation on mangrove-class F1). Hyperparameter tuning yielded only a marginal change in independent GMW v3 accuracy relative to the untuned baseline. The persistent gap between the very high cross-validation F1 (0.962) and the more modest GMW v3 F1 (0.615) cannot be closed by optimization, indicating that pseudo-label quality sets the accuracy ceiling rather than classifier capacity. This directly situates the pipeline within the label-optimization perspective of Hauser et al. (2025) and directs future efforts toward label refinement instead of further tuning.

4. Results

4.1 Sangatta training accuracy

Evaluated against GMW v3 (2020), the XGBoost Tuned model reaches Overall Accuracy 0.887, Cohen's Kappa 0.548, Precision 0.625, Recall 0.605, F1 0.615, and IoU 0.444; the untuned baseline gives near-identical agreement (Kappa 0.551) (see **Table 2**), confirming the pseudo-label ceiling. The model was trained on 8,249 pseudo-label mangrove pixels within a 57,014-pixel candidate zone. The high cross-validation F1 (0.962) against the modest GMW v3 F1 (0.615) reflects two compounding factors rather than overfitting. First, an internal circularity (pseudo-labels derive from the same spectral data used for prediction). Second, a temporal mismatch between GMW v3 (2020) and the Tanager scene (2025), which together make precision and Kappa conservative estimates.

Table 2. Confusion matrix for Sangatta (XGBoost Tuned) vs GMW v3 (2020) within the coastal candidate zone (57,014 pixels). TP true positive, FP false positive (commission), FN false negative (omission), TN true negative.

XGBoost Tuned vs GMW v3	Reference: Mangrove	Reference: Non-mangrove	Total (predicted)
Predicted: Mangrove	5,146 (TP)	3,091 (FP)	8,237
Predicted: Non-mangrove	3,360 (FN)	45,417 (TN)	48,777
Total (reference)	8,506	48,508	**57,014**

4.2 Zero-shot transferability, transfer-site results and spatial behavior

El Salvador (Kappa 0.579, F1 0.826) and Australia (Kappa 0.627, F1 0.783) matched or exceeded the training baseline, the strongest evidence that the framework generalizes across geographically, climatically, and ecologically distant mangrove systems (**Table 3**). The two Belize scenes retained high recall (0.807, 0.732) but lost precision (0.421, 0.178), indicating reliable detection of genuine mangrove alongside over-prediction into surrounding coastal cover. Notably, predicted mangrove is not confined to the shoreline as found at Belize and Australia where the positive predictions extend inland (**Figure 2**). This is an expected consequence of two design choices. First, the candidate zone is grown from every MNDWI-detected water pixel, including tidal creeks, rivers, lagoons, and interior flooded surfaces, so pixels up to 500 m from any inland water remain eligible. Second, the nine features are purely spectral and carry no salinity or tidal-inundation information, so the classifier cannot separate salt-tolerant mangrove from spectrally similar freshwater or riparian vegetation.

Table 3. XGBoost model performance in all scenes.

Site	Kappa	Precision	Recall	F1	Note
Sangatta (train)	0.548	0.625	0.605	0.615	Baseline
El Salvador	0.579	0.943	0.734	0.826	Model applied successfully
Australia	0.627	0.855	0.722	0.783	Model applied successfully
Belize 1	0.505	0.421	0.807	0.553	Over-predict (moderate)
Belize 2	0.236	0.178	0.732	0.287	Over-predict (edge/low ref)
Gujarat	−0.001	0.000	0.000	0.000	Complete failure

At El Salvador predictions remain concentrated in the estuarine delta (Precision 0.943), with minimal inland leakage. Accuracy remains higher in Australia (0.855), even though the mapping follows a network of tidal tributaries branching far inland. This is due to macrotidal conditions, where mangroves colonize the banks of upstream tidal channels. Consequently, most inland pixels align with GMW v3 and do not represent errors. At Belize the inland predictions are scattered across a low-relief karst-and-wetland matrix with abundant freshwater wetlands and residual cloud-shadow, and are predominantly commission errors, the spatial signature of the over-prediction quantified above. This behavior is the spatially explicit form of the mangrove and non-mangrove misclassification documented by Munawaroh et al. (2025), consistent with Lassalle et al. (2023) on the weakening of spectral separability where surrounding wet vegetation shares the same contrast, and with Nie et al. (2026) on the poor generalization of index-threshold methods off the calibration distribution. Addressing these misclassifications highlights a field-data-free refinement path that restricts the water anchor to tidally connected water, adds a DEM-based tidal-inundation constraint, and applies a post-classification connectivity filter.

Transfer failed completely at Gujarat (Kappa -0.001, Recall 0.000). None of the 2,565 GMW v3 mangrove pixels within the candidate zone were recovered. The root cause is extreme sparsity, as mangroves constitute only 1.87% of the candidate zone compared to the 4.1 to 66.9% density range seen at all other transfer sites. This finding is corroborated by (Nie et al., 2026), showing that the adaptive-threshold framework requires the target index distribution to be at least weakly bimodal. When the mangrove population is this small it forms no distinguishable mode, and the threshold is placed within the dominant non-mangrove distribution, effectively excluding the sparse mangrove signal.

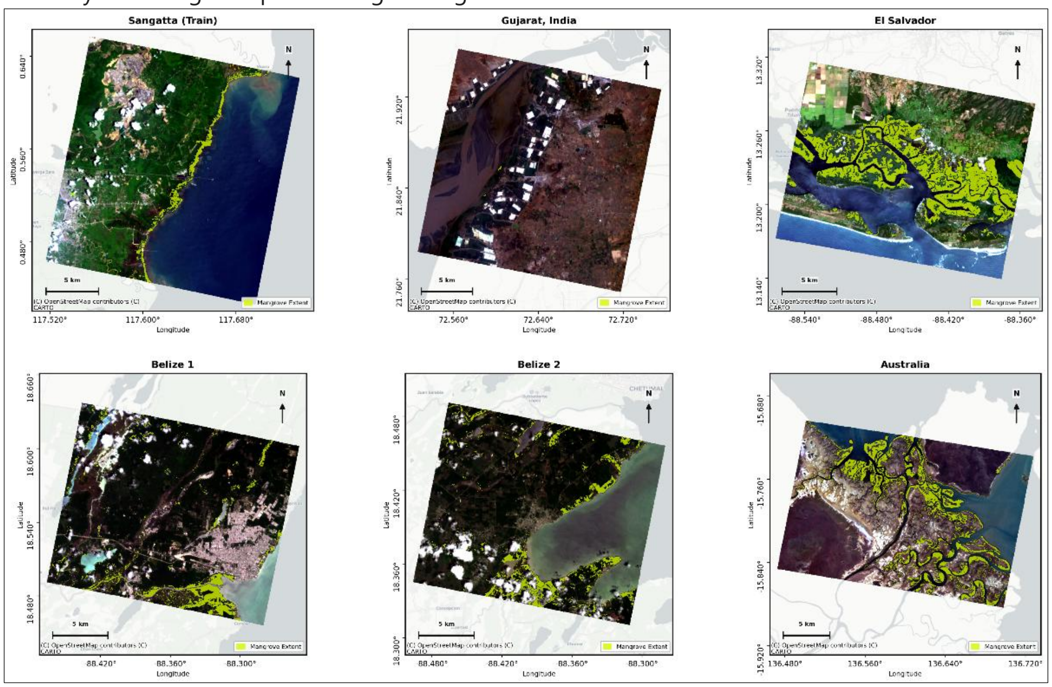

Figure 2. Multi-site mangrove predicted extent.

5. Limitations

- Pseudo-label quality sets the accuracy ceiling because the gap between cross-validation F1 (0.962) and GMW v3 F1 (0.615) reflects an internal circularity that model tuning cannot close. Label-purity improvements (spectral filters, GEDI-assisted verification) are likely to help more than any model-side change.
- Extreme mangrove sparsity breaks the adaptive-threshold assumption and produces complete failure, as at Gujarat (mangrove 1.87 % of the candidate zone). The method's operational envelope is scenes where mangrove forms a spectrally coherent, non-trivial fraction of the coastal zone.
- The temporal mismatch between the 2020 GMW v3 reference and 2025 scenes scores genuine five-year changes as errors, downwardly biasing precision and making the reported metrics conservative.
- The lack of salinity or tidal context causes inland water anchors to misclassify freshwater riparian vegetation as mangroves. Furthermore, standard multispectral deployments would drop the REIP feature because it requires contiguous hyperspectral red-edge bands.

6. Future Work

- Extending the current extent product to include above-ground biomass and blue-carbon mapping via GEDI L4A fusion, which is already confirmed available for the Sangatta scene.
- Following by Lassalle et al. (2023),future iterations will incorporate candidate-zone refinement (tidally connected water anchors, DEM-based masking, connectivity filtering) and spectral unmixing to enhance transfer at sparse and spectrally mixed sites.
- Conducting temporally matched validation using up-to-date field references to resolve the 2020-to-2025 mismatch.

7. Data and Code Availability

All Tanager-1 imagery was obtained from the Planet Open STAC catalogue as atmospherically corrected surface reflectance (HDF5). All source code and the three processing notebooks (NB01 to NB03) are released under the MIT License at github.com/mwahyur46/tanager-mangrove-mapping, with a conda environment for reproducibility. Global Mangrove Watch v3 (2020) subsets (globalmangrovewatch.org/) served as the independent accuracy reference only.

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